

## Project Initialization and Planning Phase

|               |                                 |
|---------------|---------------------------------|
| Date          | 03 July 2024                    |
| Team ID       | SWTID-2026-9778                 |
| Project Title | Credit Card Approval Prediction |
| Maximum Marks | 5 Marks                         |

### Project Proposal (Proposed Solution) report

The proposal report aims to transform the **credit card approval process** using machine learning, improving efficiency, accuracy, and consistency in decision-making. It addresses delays, manual verification, and incorrect approval decisions while ensuring faster processing, reduced financial risks, and enhanced customer satisfaction. Key features include a machine learning-based credit card approval model and real-time decision support.

| Project Overview  |                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective is to improve the credit card approval process by implementing advanced machine learning techniques that accurately predict whether a credit card application should be approved or rejected. The system aims to reduce manual effort, increase decision accuracy, minimize processing time, and assist financial institutions in making reliable credit decisions.                          |
| Scope             | The project focuses on developing a machine learning-based credit card approval prediction system that analyzes applicant information such as income, employment status, credit history, financial obligations, and demographic details. The system includes data preprocessing, feature engineering, model training, prediction, and evaluation to support efficient and accurate credit card approval decisions. |
| Problem Statement |                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Description       | Traditional credit card approval systems rely heavily on manual verification and predefined rules, which often result in slow processing, inconsistent decisions, and human errors. These limitations reduce operational efficiency and may lead to incorrect approvals or rejections, affecting both financial institutions and customers.                                                                        |
| Impact            | Solving these issues will improve approval accuracy, reduce processing time, minimize financial risks caused by incorrect decisions, enhance customer satisfaction through faster responses, and support financial institutions in making fair, reliable, and data-driven credit card approval decisions.                                                                                                          |
| Proposed Solution |                                                                                                                                                                                                                                                                                                                                                                                                                    |

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|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Approach     | Develop a machine learning-based prediction system that analyzes applicant information, performs data preprocessing and feature engineering, trains predictive models, and accurately determines whether a credit card application should be approved or rejected. The system continuously improves prediction performance through model evaluation and optimization. |
| Key Features | - Implementation of a machine learning-based credit card approval prediction model.                                                                                                                                                                                                                                                                                   |

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|  | <ul style="list-style-type: none"> <li>- Automated data preprocessing and feature engineering.</li> <li>- Real-time prediction of credit card approval status.</li> </ul> |
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## Resource Requirements

| Resource Type           | Description                             | Specification/Allocation                                                                                    |
|-------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Hardware</b>         |                                         |                                                                                                             |
| Computing Resources     | CPU/GPU specifications, number of cores | Intel Core i5/i7 Processor or T4 GPU                                                                        |
| Memory                  | RAM specifications                      | 8 GB(minimum)                                                                                               |
| Storage                 | Disk space for data, models, and logs   | 1 TB SSD                                                                                                    |
| <b>Software</b>         |                                         |                                                                                                             |
| Frameworks              | Python frameworks                       | Flask                                                                                                       |
| Libraries               | Additional libraries                    | Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost                                                   |
| Development Environment | IDE                                     | Jupyter Notebook, VS Code, PyCharm                                                                          |
| <b>Data</b>             |                                         |                                                                                                             |
| Data                    | Source, size, format                    | Kaggle Credit Card Approval Dataset, 970+ records, CSV<br><br>UCI Credit Approval Dataset, 690 records, CSV |